

Bengali Hate Speech Detection with BERT and Deep Learning Models



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Abstract An increasing amount of harmful effects have been linked to prolonged exposure to abusive language on numerous social media sites. If we want to keep the internet safe and peaceful, we must do something about the epidemic of harsh language. Although studies on the topic of identifying hostile speech have been conducted, the vast majority have only covered the English language. Recent instances in Bangladesh, however, have led to the emergence of inflammatory speech in a variety of languages. Therefore, it is crucial to address this type of harmful material. Unfortunately, Bangla hate speech detection on social media sites such as Facebook and YouTube has been hampered by a lack of available public Bangla datasets. Although some datasets are available online, they are sparse, poorly sequenced, and lack necessary data types. As a means of filling this void, we have compiled a new dataset consisting of 8600 user comments from Facebook and YouTube, which we have divided into the following five categories: sports, religion, politics, entertainment, and others. Following that, we used five distinct models to perform a massive study of abusive language in Bengali. After testing a number of different models, we found that the BERT model had the highest accuracy of 80%. The availability of this dataset greatly aids our contribution to the study of identifying hate speech in Bengali. The same models have also been run on an existing dataset of 30,000 records, where we achieved an accuracy of 97%.

Keywords Bengali hate speech detection · Deep learning methods · Custom CNN · RNN · Transformer

1 Introduction

The proliferation of social media and online forums has led to a rise in hate speech, which may have devastating effects on individuals and communities [12, 19]. Keeping the internet welcoming and secure for all people requires hate speech detection in

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a variety of languages. While research into identifying hate speech in high-resource languages has progressed, the same has not been true for low-resource languages such as Bengali. According to [26], the seventh most spoken language in the world is Bengali and has a sizable online community. A major obstacle to identifying hate speech in this language is the dearth of annotated data and linguistic resources. By facilitating the identification and removal of hateful content from online platforms, a safer environment can be created for users, and the development of an effective Bangla hate speech detection system can contribute to this goal.

Since the development of artificial intelligence, there have been a lot of different approaches that have been tried out in order to solve different kinds of problems. Learning algorithms such as machine learning and deep learning have seen widespread application in recent years for the purpose of resolving a wide variety of classification and recognition problems [6, 16]. In this research, we discuss results of a study that we conducted utilizing deep learning and transformer-based techniques to identify instances of hate speech in Bangla. We amassed a new dataset consisting of around 8600 user comments from social media platforms Facebook and YouTube. Our collection of data features Bengali texts that both contain and do not contain examples of hate speech. We have divided all of our information into five distinct categories, which are as follows: entertainment, religion, politics, sports, and others. We have used a variety of models in order to achieve the best outcome possible. These models are LSTM [29], Bi-LSTM [14], GRU [3], CNN [18], and BERT [11]. The BERT model, which maintains 80% accuracy on test data, is the model that has fared the best out of these five models. Based on the findings of our research, it is clear that the suggested model is capable of achieving a high level of accuracy when identifying hate speech in the Bengali language.

In addition to this, we have employed the very same models to carry out analyses on an existing data collection that is comprised of 30,000 records [24]. The author of the study applied three different models to that dataset (SVM, LSTM, and Bi-LSTM), and as a result, they achieved an accuracy of 87.5%. However, when we put our models into practice, we achieved an accuracy of more than 90% across all of our models. By utilizing the BERT model, we were able to get the highest level of accuracy, which was 97%.

The following is an illustration of our substantial contributions to this work:

1. We compiled a fresh set of 8600 records and classified them as hate speech or non-hate speech. It is further broken down into the religious, entertainment, political, sports, and other sub-genres. The information is gathered from several online platforms.
2. Examine several deep learning and transformer models using our dataset and compare their results.
3. Evaluate the performance of our models using the existing dataset.

2 Related Research

There has been a significant amount of research conducted in the past to identify hate speech via deep learning models [7, 24], but only a small number of studies have employed transformer-based approaches. The subsequent descriptions encompass prior research conducted on the Bangla language as well as other linguistic domains.

2.1 *Related Research on Other Languages*

Early efforts to establish resources for identifying hate speech concentrated primarily on the English language [1, 9, 28]. In order to make multilingual hate speech datasets, several shared task competitions have been held recently, such as HASOC [20], OffensEval [30], TRAC [17], and many datasets, such as Hindi [22], Danish [25], Greek [23], Turkish [4], Mexican Spanish [2], etc., have been made. Work is also being done to figure out how to identify hate speech in Bangla.

A lot of the early research on abusive language focused on extracting lexical and syntactic elements from online messages in an effort to identify abusive people. The field of automated hate speech identification has come a long way in the past few years, thanks to much academic study. Davidson et al. [10] worked on differentiating offensive materials from hateful material on Twitter. The authors used a wide range of classifiers (including LR, SVM, and decision trees) on this dataset to investigate n -grams, emotion lexicons, POS tags, and TF-IDF vectors. In 2016, Zeerak et al. [28] added a dataset where thousands of messages were classified as racism and sexism markers. Even though many datasets were made available, the lack of quality and completeness remained a significant issue.

2.2 *Related Research on Bangla Language*

Recently, it has been challenging to work on large-scale hate detection projects in Bengali due to a lack of adequate datasets. While there are some datasets available online, they are sparse, poorly sequenced, and missing essential data types. However, only a handful of papers have been published in recent years, including [24]. Thirty thousand Bengali comments from YouTube and Facebook, including ten thousand pieces of hate speech, make up their data collection. Several deep learning models, including Word2Vec, FastText, and pretrained BengFastText, have been run. All deep learning models performed well; however, the experiment showed that SVM had the highest accuracy (87.5%).

Ismam et al. [15] conducted a study wherein they gathered and annotated 5000 comments from Facebook. These comments were categorized into six distinct groups, namely religious hate, inciteful comments, political comments, hate speech, religious

Table 1 Articles summary based on hate speech detection using Bengali language

Authors	Methods	Accuracy (%)
Das et al. [8]	MuRIL, XLM-roberta, indicBERT, m-BERT	MuRIL 90.6
Romim et al. [24]	SVM, LSTM, Bi-LSTM	SVM 87.5
Kumer et al. [5]	CNN, GRU, LSTM, attention-based decoder	Attention-based decoder 77
Ismam et al. [15]	LinearSVC, random forest, Naive Bayes, GRU	GRU 70.10

comments, and communal attack. The author employed several machine learning algorithms and attained 70.10% accuracy in GRU-based model. All of the most recent articles containing Bengali hate speech are summarized in Table 1.

BERT and XLMRoBERTa, both of which are transformer-based language models [27], are gaining popularity in a number of downstream tasks. Several previous deep learning models have been shown to be inferior to those based on transformers, as has already been observed [21]. Since we have found that transformer-based models perform best on our classification task, we will be focusing on developing such models. The BERT model, which we employed after considering the widespread adoption of transformer-based models, has produced impressive results on our data.

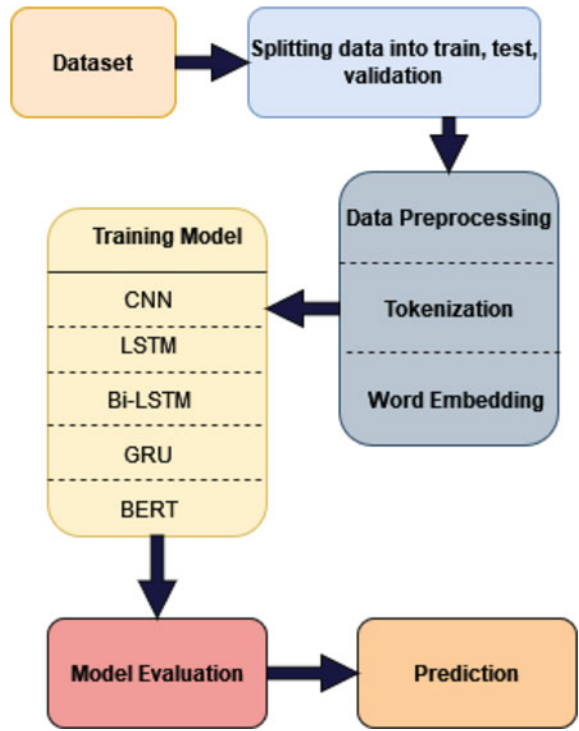
3 Materials and Proposed Methodology

In this part, we laid out the methodology we recommend for recognizing hate speech in Bengali. First, a new training dataset is produced by collecting data from social media networks like Facebook and YouTube and storing it as a comma-separated values (CSV) file. About 70% of the data was utilized for training, 15% of the data for validation, and 15% of the data for testing purposes. Keras preprocessing techniques were used to get control of the data. To evaluate the efficacy of several unique classification methods, we tested CNN, LSTM, Bi-directional LSTM, GRU, and BERT. After exhaustively comparing and contrasting various methods of classification, we have arrived at an accurate prediction. Our method’s workflow is depicted in Fig. 1.

3.1 Architecture of the Models

Five models are trained using characteristics from our dataset, and the results are compared to see which one is the most effective at detecting hate speech in Ben-

Fig. 1 Workflow of our approach



gali. In our experiment, we tested CNN, LSTM, Bi-LSTM, GRU, four of the most well-known deep learning models, and the One Transformer Model (BERT). The architecture of those models is described further down in this section.

Convolution Neural Network (CNN): For the CNN model, we have experimented with a variety of unique layer combinations. Based on our findings, the optimal configuration for a CNN model is 100 units of CNN convolution layers along with max-pooling as well as 64, 32, and 16 units of dense layers. We have used Adam as the optimizer and chosen a learning rate of 0.001 and a dropout of 0.2. As a loss function, we have also employed binary cross-entropy.

Long Short-Term Memory (LSTM): In our experiment, we employed 300 LSTM layers, split between 128 and 64 dense layers. We have chosen a learning rate of 0.001 and an optimizer of RMSprop with a dropout of 0.2. Mean squared error (MSE) has also been utilized as a loss function.

Bi-Directional Long Short-Term Memory (Bi-LSTM): We trained with 128 units of backward and forward LSTM, 64 and 32 units of dense layers, a dropout rate of 0.2, and a learning rate of 0.0002 using the RMSprop optimizer. We have chosen mean squared error (MSE) as our loss function.

Gated Recurrent Units (GRU): We have deployed 32 GRU layers at a dropout rate of 0.2 per unit and 16 dense layers. The Adam optimizer was employed. In this case, we choose mean squared error (MSE) as our loss function and a learning rate of 0.0002.

Bi-directional Encoder Representations from Transformers (BERT): Previous studies have demonstrated that transformer models, trained in monolingual, multilingual, or cross-lingual settings, have attained the most advanced performance in the classification of objectionable texts. Consequently, we employ a multilingual Bi-directional Encoder Representations from Transformers (m-BERT) model that is trained on 104 languages sourced from the extensive Wikipedia dataset. The model consists of a stack of transformer encoder layers, each containing 12 attention heads. These encoder layers are fully connected neural networks that incorporate a self-attention mechanism. There is a cap on the number of tokens that can be used with m-BERT, which currently sits at 512. A completely linked layer, where the output is the CLS token from the input, is also added to m-BERT for fine-tuning purposes. Typically, the model's representation of the sentence is contained in the CLS token output. After extensive research into abusive language, the m-BERT model has proven superior to the standard of excellence. Our proposed BERT architecture, which outperformed all used custom models, is depicted in Fig. 2. The program executes a total of 10 iterations using the Adam optimizer. The batch size is set to 16, the learning rate is $2e-5$, and the Adam epsilon value is $1e-8$. For our experiments, a token size of $n = 128$ was chosen.

3.2 Performance Evaluation

In this study, the models undergo training, validation, and testing using distinct sets of data. Models are trained using examples from the training dataset, while hyperparameters are chosen from the validation dataset. Models are assessed using a number of statistical measures using test data from which they have never been exposed to determine how well they perform. The effectiveness of the models is measured with tools like accuracy, recall, precision, $F1$ -score, and confusion matrix. One common metric for evaluating the efficacy of a classification algorithm is the confusion matrix, which provides a graphical summary of the model's results. Confusion matrices display the proportion of correctly categorized positive and negative examples as true positives (TP) and true negatives (TN), respectively. On the other hand, the quantities of false positives (FP) and false negatives (FN) indicate the ratio of correctly identified positive and negative instances that were inaccurately classified. The confusion matrix has four factors used to determine metrics like accuracy, recall, precision, and $F1$ -score. Values between 0 and 1 represent the quality of the classifier, with 1 being the best and 0 the worst.

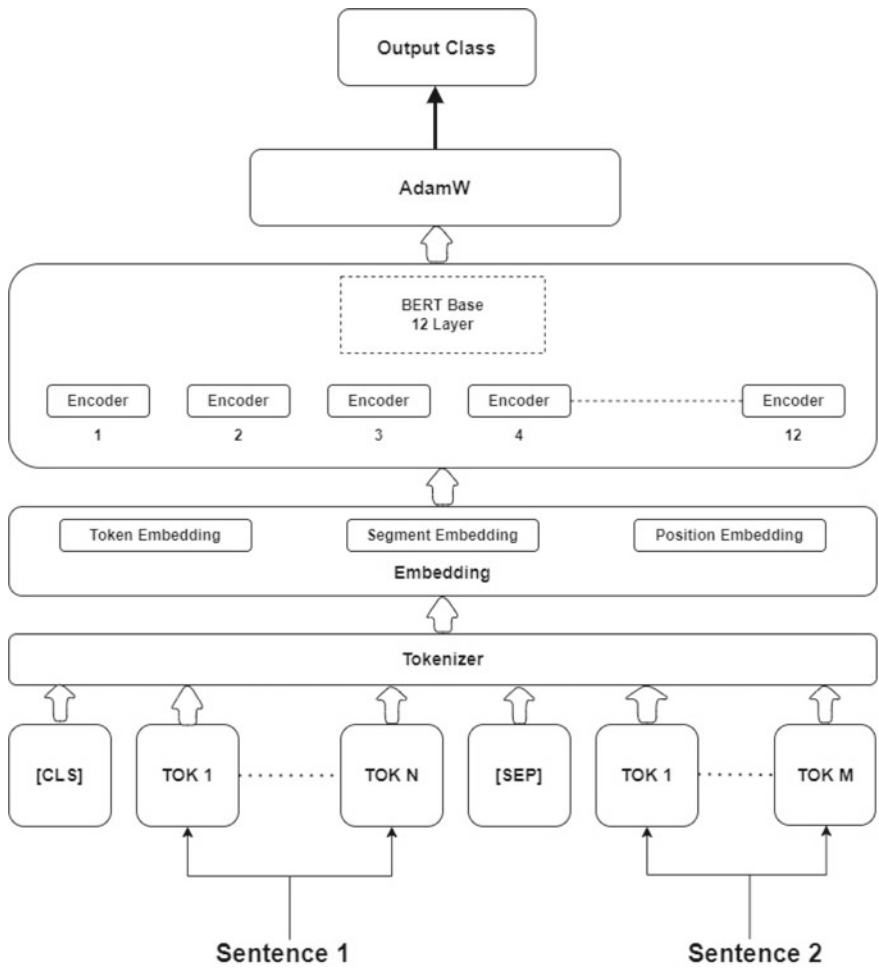


Fig. 2 Architecture of our BERT model

4 Result and Observation

Here, we provide experimental results and a comprehensive performance analysis of our approaches to hate speech classification in Bangla. The results of the various models, along with a discussion of how they stack up against other available datasets, follow below. Our tests were conducted on the Google Collaborator platform using the GPU back-end for Python 3 on the Google Cloud. To clean and organize the data, we utilized the latest versions of pandas (1.1.4) and NumPy (1.18.5). We use Keras (2.4.0) and TensorFlow (2.3.0) to construct our deep learning models.

4.1 Dataset

The effectiveness of a model is proportional to the quality of the data used to train it. Therefore, gathering relevant and useful data is a crucial first step in the model-building process.

Data Collection: According to a recent statistical analysis, Facebook and YouTube have the largest combined user bases of all social networking sites [12]. Around 2.95 billion people are active monthly Facebook users, sharing and connecting in all corners of the globe. Considering its widespread use, the vast majority of our information came from the comments sections of Facebook and YouTube posts. We have amassed information from a variety of celebrity pages, including the most popular posts and comments on the most popular videos. Creating a wide range of data types was a top priority during the dataset's development. As a result, we culled feedback from Facebook and YouTube in five distinct subject areas: arts and culture, politics, sports, faith, and society. The news websites Prothom Alo, Dainik Azadi, and Dainik Purbokone were mined for their religious statistics. We have data from the most recent trending photographs and videos on Facebook and YouTube for various fields, such as sports and politics.

Data Annotation: When used to characterize harmful communication that targets a group or a person based on their identity (such as their race, gender, or religion), the phrase "hate speech" is commonly employed [19]. The United Nations' Strategy and Plan of Action on Hate Speech provides a global framework for combating this pervasive problem. Hate speech is defined by the United Nations as any type of communication that targets individuals or groups on the basis of their religion, nationality, ethnicity, race, descent, skin color, gender, or other identification factor and employs disparaging or discriminating terminology [13]. We have collected our statistics on hate speech in accordance with UN standards. We have also adopted the annotation strategy used by other articles in the available research [5, 8, 24].

Data Statistics: The volume of data in each category will be discussed in this section of the paper. We have gathered a total of 8600 data, 53% of which are not hate speech and are labeled as 0 and 47% of which are hate speech and are labeled as 1. We get a deeper comprehension with the aid of numerical examples, as shown in Fig. 3.

The religion category has the most records in this data collection. There are 1737 items in this group. In addition, there are 1673 entries in the entertainment area, 1630 in politics, and 1590 in sports. The others category has 1515 records, the fewest of any in the developed corpus. All of our five types of data have been further divided between "non-hate speech" (category 0) and "hate speech" (category 1). The overall distribution of categories is shown in Fig. 4.

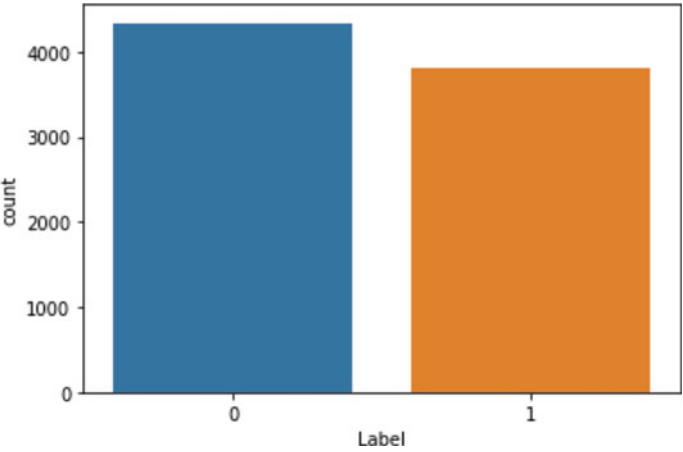


Fig. 3 Distribution of 0 (non-hate speech) and 1 (hate speech) in our dataset

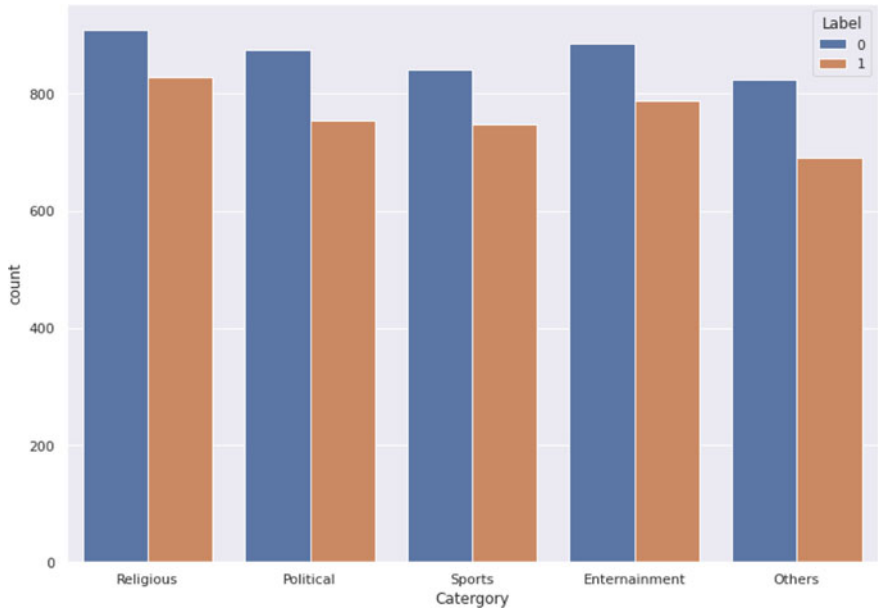


Fig. 4 Distribution of non-hate (0) and hate (1) speech based on category

4.2 Data Preprocessing

We have processed the raw text further after gathering all of the comments, removing all of the punctuation, mentioning, bad characters, etc. Our preprocessing system consists of the following components:

Before preprocessing

“শুয়োরে বাচ্চারা তোদের খাইয়া কাজ নাই। এ দেশের যুবসমাজকে কোনদিকে নিয়ে যাচ্ছি। তোদের কোন দায়িত্ব নাই? পরিমনির হল কার্তিক মাসের কুন্তি আর এর পিছনে তোরা সবকটা কুন্তা। 🙄🙄🙄”

After preprocessing

“শুয়োরে বাচ্চারা তোদের খাইয়া কাজ নাই এ দেশের যুবসমাজকে কোনদিকে নিয়ে যাচ্ছি তোদের কোন দায়িত্ব নাই পরিমনির হল কার্তিক মাসের কুন্তি আর এর পিছনে তোরা সবকটা কুন্তা”

Fig. 5 Data before and after preprocessing

Fig. 6 Sentence before and after tokenization

before tokenization

“কুন্ডায় ও খাইবেনা ওর পঁচা দেহ”

after tokenization

[‘কুন্ডায়’, ‘ও’, ‘খাইবেনা’, ‘ওর’, ‘পঁচা’, ‘দেহ’]

Removal of Emoji, Punctuations, and Non-Bengali Language: The unprocessed Bengali comments that we used to compile our dataset include texts that use emoji, punctuation, and the English alphabet. Here, we have cleaned up the text by removing all the emoji, numbers, non-Bengali letters, symbols, and punctuation that didn’t belong. All the undesirable characters have been erased throughout this procedure. A comparison of raw data and data after preprocessing is shown in Fig. 5.

Tokenization: Tokenization is a straightforward method for transforming an unstructured dataset into something more manageable. Paragraphs and sentences are broken down into clauses and phrases so that meaning may be more precisely applied. The second stage was to tokenize the original Bengali text. A text is shown before and after being tokenized in Fig. 6.

Word Embedding: We have utilized the Keras embedding layer to incorporate words in our experiments. With the help of the embedding layer, we may transform each word into a vector of a certain length. Each word may be embedded in a vector of a predetermined length, thanks to Keras’s embedding layer. While the one-hot-encoding technique generates a massive sparse matrix to represent a single word, embedding layers allocate a real-valued vector of a defined length to each word. One of the benefits of this approach is that it requires fewer dimensions per word to be conveyed. Word representations using this method use much less space than those using one-hot encoding because of the reduction in dimensionality.

Hyperparameters of Our Proposed Model: During the process of constructing the BERT model that we presented, we put our model through 10 iterations using the Adam optimizer. We discovered that the BERT model with the hyperparameters that are presented in Table 2 generated the most favorable outcomes.

Table 2 BERT model hyperparameter setup

Optimizer	Epsilon	Learning rate	Batch size
AdamW	1e−8	2e−5	16

Table 3 Summary of several model’s performances on our dataset

Model	Label	Precision	Recall	F1-score	Accuracy
CNN	NHS (class 0)	78	79	79	77
	HS (class 1)	75	73	74	
LSTM	NHS (class 0)	79	79	79	77
	HS (class 1)	75	75	75	
Bi-LSTM	NHS (class 0)	79	78	78	77
	HS (class 1)	75	75	75	
GRU	NHS (class 0)	78	76	77	78
	HS (class 1)	73	75	74	
BERT	NHS (class 0)	82	82	82	80
	HS (class 1)	79	79	79	

4.3 Performance Analysis

Models Performance on Our Dataset Through model experimentation, we have tried various configurations of layers and hyperparameters in an effort to converge on a generalized model with good performance on unseen instances and to determine the optimal values for the hyperparameters. In Table 3, we compare the five models we employed and summarize their performance on test data. Based on the performance assessment matrix displayed in Table 3, we have only taken into account the best possible results for each model. The test set is where we report all of our findings. With an accuracy of 80%, the BERT model outperformed the bespoke deep learning models. Figure 7 and Fig. 8 depict the proposed BERT model’s accuracy and loss curve, respectively. Our model’s confusion matrix is shown in Fig. 9.

Models Performance on Existing Dataset We have also evaluated our models on a similar existing dataset [24] to see how it does on additional datasets. There are a total of 30,000 recordings, and they have been divided up into seven distinct categories. The vast majority of these data originate from social media sites like Facebook and YouTube, and 10,000 of them contain examples of hate speech. Our models performed better than the author’s models, as mentioned in Table 4.

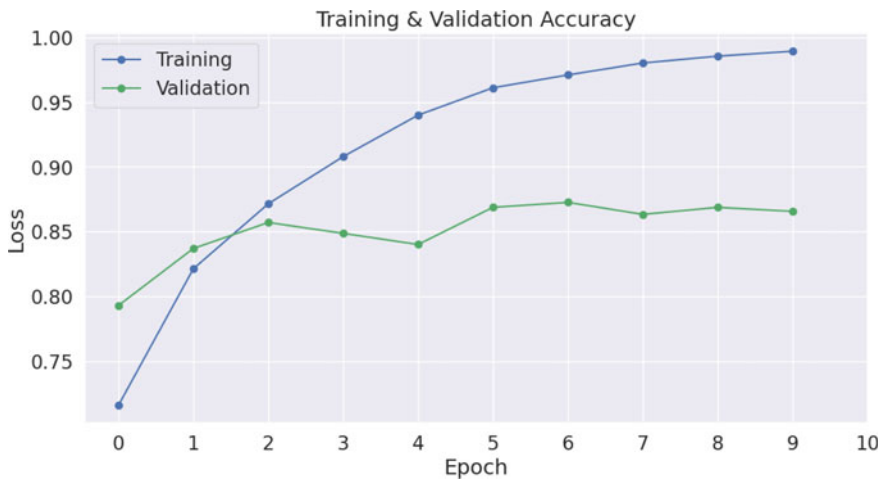


Fig. 7 Accuracy curve of our proposed BERT model

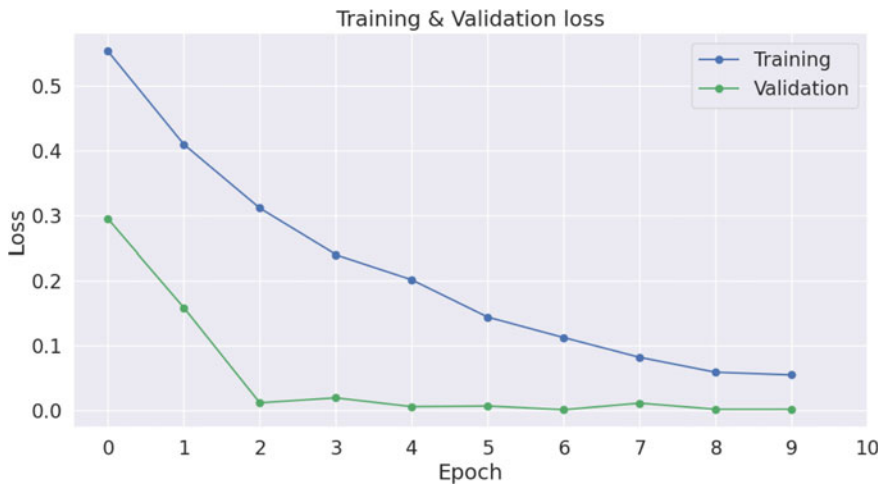


Fig. 8 Loss curve of our proposed BERT model

5 Conclusion and Future Work

In this research, we provide a comparative analysis of abusive speech in Bengali as a means of combating the growing prevalence of abusive and nasty language on online social media. We provided an informational dataset consisting of 8600 Facebook and YouTube user comments that were split into five distinct groups. The BERT model achieved an accuracy of 80% on the test dataset, which was the highest of the five

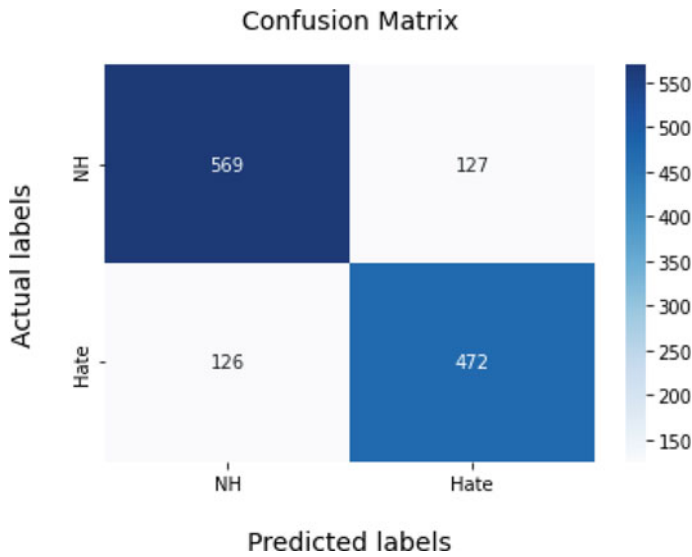


Fig. 9 Confusion matrix of our proposed BERT model

Table 4 Comparison of our models on an existing dataset

Authors	Model	Accuracy	F1-score
Romim et al. [24]	LSTM	84.3	89
	Bi-LSTM	86.55	90.1
Ours	CNN	93	95
	LSTM	94	96
	Bi-LSTM	93	95
	GRU	95	96
	BERT	97	96

customized models we tested (CNN, LSTM, Bi-LSTM, GRU, and BERT). We also used the same models on an existing dataset of 30,000 records. All of our models had an accuracy of 90% or above, and the BERT model was the most accurate, with a 97% success rate and an *F1*-score of 96%, surpassing the current accuracy documented in the existing research work by 5%. To further extend our dataset, we hope to incorporate additional information on a wide range of issues in the future. Our models will undergo an effort to enhance their performance.

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